3.29 Local Rings of Dimension 0

Lemma 3.29.1 (Dimension 0 local ring)
Let $(A, \mathfrak{m})$ be a Noetherian local ring. Then the following are equivalent

a) $\dim A = 0$ *(i.e. every prime ideal is maximal)*
b) $\mathfrak{m}^n = 0$ *for some* n
c) $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ *for some* n

If A is an integral domain then this is equivalent to A being a field.

Proof. a) $\implies$ b) By (3.22.6) $\mathfrak{r} := \sqrt{(0)} = \mathfrak{p}_1 \cap \ldots \cap \mathfrak{p}_n$ is a decomposition into minimal prime ideals. By assumption these are also maximal, and therefore by uniqueness $\sqrt{(0)} = \mathfrak{m}$ and the result follows from (3.3.47) as $\mathfrak{m}$ is finitely generated.

b) $\implies$ a) Let $\mathfrak{p}$ be a prime ideal then trivially $\mathfrak{m}^n \subset \mathfrak{p}$ so by (3.28.1) it is $\mathfrak{m}$-primary i.e. $\mathfrak{m}$ is a minimal prime of $\mathfrak{p}$ whence $\mathfrak{m} = \mathfrak{p}$.

b) $\iff$ c) One direction is obvious, and the converse follows from Nakayama's Lemma (3.21.5) because $\mathfrak{m}^n$ is finitely generated. ∎